

Technical Drawing with AutoCAD Electrical - Hands-On Labs

WSQ Course Code: TGS-2023037466 | Conducted by: Tertiary Infotech Academy Pte Ltd (UEN 201200696W) | Version v6.1

Each lab lives in its own folder with the instructions as Markdown (.md) and PDF, plus the starter AutoCAD DWG file(s) used in the lab. Complete the labs in order - they build on each other within each topic.

Topic 01 - Introduction to Technical Drawing Using AutoCAD Electrical

- [Lab 1: Set Up a New AutoCAD Electrical Drawing](#) - A correctly configured metric drawing sheet, saved as a reusable baseline for the rest of the course.
- [Lab 2: Create a Project with Project Manager](#) - A verified working copy of an AutoCAD Electrical .wdp project, with its schematic and panel sheets in the correct order and current library paths resolved.

Topic 02 - Analysing Technical Drawings

- [Lab 3: Build Circuits with Circuit Builder](#) - A schematic sheet containing a Circuit Builder motor circuit plus a saved, reusable circuit block.
- [Lab 4: Insert Wires, Wire Numbers and Terminals](#) - A correctly wired and wire-numbered circuit with terminals, matching the drawing's wire-number specification.
- [Lab 5: Cross-Reference and Surf a Drawing Set](#) - A cross-referenced two-sheet circuit you can navigate with Surfer, plus a Mark/Verify change report.

Topic 03 - Reviewing Technical Drawings for Strategic Relevance and Standard Compliance

- [Lab 6: Create a Panel Layout with Footprints](#) - A panel layout drawing with footprints placed from the schematic list, a nameplate, item numbers and balloons.
- [Lab 7: Generate Reports and Part Lists](#) - A BOM table placed on the drawing plus a saved .SET format file and a CSV export of the report.
- [Lab 8: Plot and Publish the Project to PDF](#) - An A3 PDF plot of one sheet and a hyperlinked multi-sheet PDF of the whole project.

DWG starter-file sources

The starter drawings are reused with thanks from these open exercise-file repositories:

- [DCS-training/AutocadSelfLearn](#)
- [PacktPublishing/AutoCAD-2025-Best-Practices-Tips-and-Techniques](#)
- [PacktPublishing/Mastering-AutoCAD](#)

Selected labs also include an instructor-supplied AutoCAD Electrical sample project in the controlled Google Drive course-delivery package. Those project files are used on copies for realistic project, schematic, panel, report and publish workflows, and are intentionally excluded from the public GitHub repository.